Sweatshop on P-Chess: lane `mvp`

CI test job, themes and sounds, the engine fetch script (CI-001, UI-003, BOT-001) and the Windows export (CI-002, parked); day lineup, then Opus 5.5

Foreman, session `sweatshop/mvp/2026-10-24-1310`

24 October 2026

Summary

Lane `mvp` ran in the clone `.artifacts-build/lanes/mvp`, from `main` at `66cd0ab`. It holds the `Review: human` tickets, and every queued ticket of the other lanes waited on one of them, BOT-001. Two launches: 13:10 (pid 18032, default day lineup: stage 2 `gpt-6.1-sol high`, stage 3 and hard `gpt-6.1-sol xhigh`, on Codex; launched by the coordinating session and adopted by this foreman at 13:12; paused with `STOP` at 14:09) and 14:11 (pid 5808, `-Models 'stage 2 opus-5.5 high, hard opus-5.5 xhigh, stage 3 opus-5.5 xhigh'` at Bruno's request; ended 14:16 with "Nothing runnable").

- **Produced: 3 tickets approved and merged on the session** PR #5: CI-001, UI-003, BOT-001, all `Review: human`, `Verdict: Approve`. CI-002 is implemented and reviewed on `ci-002` but parked `blocked`, by contract, outside the session.
- **Cost: $8.07** at API list price over 9 stages, all priced: `gpt-6.1-sol high` $3.50 (1 point of the weekly Codex quota), `gpt-6.1-sol xhigh` $3.14 (2 points), `opus-5.5 xhigh` $1.44 (-).
- **What went wrong:** criterion 6 of BOT-001 asked to compare the pinned hashes with Bruno's `engine/` copies, which do not exist on this machine. The first review held the ticket for Bruno; once he accepted the official GitHub artifacts as the baseline, the re-review merged in 5 minutes, but the driver then refused the reused gate receipt because its own CI-002 parking note had changed another ticket's file. The foreman reran the gate (81/81) and restored `done`.
- **Provisional verdict:** the day lineup did the work cleanly: four first-attempt handoffs, zero review findings, zero reopens. Everything that cost time was the contract (a reference that never existed) or the driver (receipt check, re-entry after a held review), not the models.

What was done

CI-001: every push now runs the test suite on GitHub Actions, on Ubuntu with Godot 4.7.2 checked against its published SHA-256, with exactly the local gate command, so a red test shows on the branch before review. Main file `.github/workflows/ci.yml`. Proof: a green push run (78/78) and, on the session tip `0b05c75`, run 40115493475, 81/81.

UI-003: the board can be themed. `ui/theme_loader.gd` loads a board `theme.json`, the twelve cburnett piece SVGs and one OGG per sound event from `res://themes/` or `user://themes/`; `ui/sound_player.gd` plays move, capture, castle, check, promotion, game start and game end. The four free Lichess sets (`sfx`, `nes`, `piano`, `futuristic`) are bundled; `sfx` is the default until Bruno picks one by ear.

BOT-001: `scripts/fetch-engine.ps1` (Windows PowerShell 5.1) downloads Lc0 v0.32.1 CPU dnnl and the nine Maia v1.0 nets into `engine/`, checks 20 pinned SHA-256 hashes (the zip, its ten entries, nine nets), deletes a file and exits 1 on any mismatch, and skips files that already verify. `docs/THIRD_PARTY.md` names both GPL-3 projects and their tags. Criterion 6 was rewritten from Bruno's decision: the hashes come from the official downloads; the Lc0 zip matches GitHub's published digest; Maia publishes none.

CI-002 (not integrated): `export_presets.cfg` and a second CI job export the Windows build and upload it as an artifact without `engine/` (49 theme assets verified in the package, run 40113823385). It waits on BOT-002 and BOT-004 for the missing-engine message (criterion 5) and on a human listening to the move sound (criterion 6).

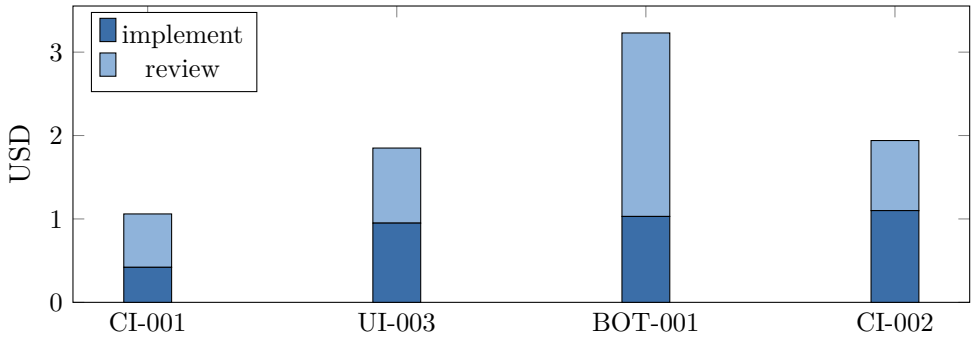
Timeline

13:10	Driver launched by the coordinating session, pid 18032, day lineup.
13:12	Adopted by this foreman. Bruno's `engine/` folder not found on disk.
13:15–13:35	CI-001 and UI-003 implemented and merged on the session, 0 findings. CI-001's stage 2 pushed `ci-001` to `origin` for its Actions proof.
13:47	BOT-001 to review. Foreman audit: downloads only from the two official releases; zip hash equals GitHub's digest.
13:51	BOT-001 review: 0 findings, held as *Needs your call*; proxy escalated criterion 6.
13:58	Bruno (relayed): accept the official artifacts as the baseline.
14:04–14:09	CI-002 implemented (pushed `ci-002`) and reviewed, parked by contract.
14:08	Bruno (relayed): stop after this ticket, switch to the Opus lineup. `STOP` honoured at 14:09; PR #5 opened.
14:10	Foreman: session revert of the parking commit (`91627ec`), decision folded into criterion 6 on `BOT-001` (`d0b0787`).
14:11–14:16	Relaunch, pid 5808. BOT-001 re-review merged and closed (`83b8d00`, `3f5e335`); driver parked it for a stale receipt (`5225e25`).
14:18	Foreman: revert (`2583e7f`), gate rerun 81/81, receipt line updated (`0b05c75`). Nothing runnable: BOT-007 waits on BOT-006, CI-003 on CI-002.

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
13:15	CI-001	impl	sol high	5	1.3M	96%	6k	0.42	to-review
13:19	CI-001	review	sol xhigh	4	1.5M	92%	10k	0.64	merged
13:31	UI-003	impl	sol high	12	2.8M	96%	20k	0.95	to-review
13:35	UI-003	review	sol xhigh	4	2.2M	91%	13k	0.90	merged
13:47	BOT-001	impl	sol high	12	2.9M	96%	22k	1.03	to-review
13:51	BOT-001	review	sol xhigh	5	1.5M	88%	15k	0.76	held for Bruno
14:04	CI-002	impl	sol high	12	3.4M	97%	21k	1.10	to-review
14:09	CI-002	review	sol xhigh	5	1.8M	90%	13k	0.84	parked (contract)
14:16	BOT-001	review	opus xhigh	5	2.7M	95%	31k	1.44	merged*
				64	20.1M		151k	8.07	

* The run-log row reads `failed (incomplete handoff), blocked`: the review merged and closed the ticket, and the driver then parked it over the receipt check. The row is left as logged; what happened is above.



No attempt was lost. BOT-001's second review ($1.44) is the price of a criterion that named a reference nobody had: the first review could only hold it. Input is 88–97% cache reads.

Comparison with previous runs

Earlier lanes today, from `run-log.md`: `skeleton` (CORE-001, $5.61), `bot` (BOT-003, $4.87 before its two reworks), `core` ($7.20 per approved ticket). Lane `mvp`: $2.69 per approved ticket ($8.07 over 3, CI-002's $1.94 included). These tickets were small (a workflow, a loader, a script), and the day lineup ran 4–12 minute implementations against 3–14 for Opus overnight.

Observations per model

- **gpt-6.1-sol high (stage 2)**: 4 runs, 4 to review on the first attempt, 5–12 minutes, $0.42–$1.10. Every handoff disclosed what it could not prove (UI-003's listening test, BOT-001's missing reference, CI-002's dependency). On both CI tickets it pushed its ticket branch to `origin` to get the Actions run the ticket demands; the foreman rule forbids pushing ticket branches.
- **gpt-6.1-sol xhigh (review)**: 4 reviews, 0 findings, 0 reopens. It held BOT-001 and CI-002 instead of approving past an open condition, and checked the Lc0 digest and Godot template hashes against upstream itself.
- **opus-5.5 xhigh (review)**: 1 re-review, 5 minutes. It reviewed only the post-review delta (ticket text), reused the gate receipt because production was unchanged, and dismissed one wrong Standards finding from its own subagent.

Driver and runtime issues

1. **The driver's own note invalidates its receipt check.** At `done` the driver diffs the tested commit against the session, excluding only the ticket and its ledger. Its CI-002 parking note (`9f61f41`) landed on the session in between, so BOT-001's valid merge was parked. Third time today (BOT-002, UI-002, BOT-001).
2. **No re-entry for a held review.** After a human answers a *Needs your call*, the session copy says `blocked`, which the driver honours over the branch. The foreman had to revert the parking commit by hand so the driver would read the branch's `to-review`.
3. **CI tickets need a push.** CI-001 and CI-002 can only prove themselves on Actions; stage 2 pushed `ci-001` and `ci-002`. Both remote branches are still on `origin`.
4. **Run-log outcome mislabels the last BOT-001 review** (above).

Recommendations

1. Exclude every `.scratch/*/issues/*.md` from the "code changed after gate" diff: ticket text is not production. Rests on three parked merges today, no scoreboard row.
2. Give the driver an `-Answer <ID>` (or document the revert) for a held review: drop the session's parking commit and let the branch's stage lead.
3. Say in `ticket-flow` how a CI ticket gets its Actions proof: push the ticket branch and delete it at merge, or run the workflow on the session branch.
4. Write criteria that name a human's local files only after checking they exist; BOT-001's criterion 6 cost one review and a relaunch.
5. Keep `gpt-6.1-sol high` → `gpt-6.1-sol xhigh` for small `Review: human` tickets: scoreboard row 89 reviews, 13% reopen, 12% S2 rate, the lowest S2 rate among pairs with more than 10 reviews.

Human debt and proxy decisions

No `Débito humano:` lines this run. `Proxy decided` lines, all by the `gpt-6-astra high` fallback:

- **BOT-001**: "Proceed with stage 2 using hashes computed from official upstream downloads; preserve comparison with Bruno's copies as an explicit human integration gate." The review's

`Proxy escalated` line sent criterion 6 to Bruno, who decided: official GitHub artifacts are the baseline.

- **CI-002**: "implement and independently review the CI-002 exporter now ... preserve criterion 5 as pending until BOT-002 and BOT-004 provide the visible missing-engine flow"; and at review, "record the completed CI-002 review with Stage: blocked on ci-002, retaining the implementation without session integration".

Reopen attribution

No reopened review this run, and no merging review authored a fix: no rows added to `.scratch/reopen-attribution.md`. PR #5 has one review, Copilot's, which did not run (quota); no session-review, release-audit or post-release finding.

Scoreboard

`scoreboard.ps1` over P-Chess (main checkout, log merged and copied back), ClassyCall, EasyMath, PhysiSyst, SIGAA-ME, SynchroNice and slip. Rows that moved since the `core` report: `gpt-6.1-sol high` → `gpt-6.1-sol xhigh` 87 → 89 reviews, 12 reopened (rate 14% → 13%), from this run's two merges (the held reviews of BOT-001 and CI-002 do not count as reviews with a verdict); `opus-5.5 high` → `opus-5.5 xhigh` 4 → 6 reviews and `opus-5.5 xhigh` → `opus-5.5 xhigh` 3 → 5, both from other lanes. Opus pairs remain under 10 reviews: anecdotes.

Implementer	Reviewer	Reviews	Reopened	Rate	S2	S2 rate
gpt-6.1-sol high	gpt-6.1-sol xhigh	89	12	13%	11	12%
gpt-6.1-sol high	gpt-6.1-sol max	94	21	22%	20	21%
opus-5.5 high	opus-5.5 xhigh	6	2	33%	2	33%
opus-5.5 xhigh	opus-5.5 xhigh	5	0	0%	0	0%

Implementers: `gpt-6.1-sol high` 216 runs, 187 to review, 8 failed, 21 asked, median 20 min, $369.21. Interventions across repos: 22 human rescues, 46 automatic recoveries, 4 routine approvals, 27 `gpt-6-astra high, fallback` proxy decisions, 15 Opus. This run's five intervention rows are committed on `main` with this report; the printout above was taken from the main checkout before that commit, so they are not in it. Full printout: the foreman's scratch output.

Pending for the human

1. Review and merge PR #5 (CI-001, UI-003, BOT-001, all `Review: human`). Every queued ticket of the other lanes waits on BOT-001 reaching `main`.
2. UI-003: listen to the four sound sets and name the default (`sfx` for now).
3. CI-002: after BOT-002 and BOT-004 are integrated, a fresh Actions artifact must show the fetch-script message on Windows, and someone must hear the move sound.
4. Delete `origin/ci-001` and `origin/ci-002`.